STA 5197 Applied Longitudinal Analysis Notes

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Preface

I don't know why this file is called index.qmd.

Here is an r code block:

```
TWO <- 1 + 1
TWO
```

```
[1] 2
```

$$\Omega = \theta x + \epsilon$$

1 ThingsToKnow

2 Things to know:

1. The design under which the data were obtained: balanced/unbalanced
2. EDA mean structure
   - Time plot
   - line plot / spaghetti plot
   - Time plot and lines for selected subjects. Selected to illustrate tracking is better than random selection.
   - Cross-sectional vs longitudinal association w/ a time-varying covariate
   - correlation structure - remove dependence on covariates first. ensures greater comparability.
   - pairs plots
   - autocorrelation for regular grid of obs times
   - variogram for irregularly spaced observations

2.1 3. Estimation

- review multiple reg. - **iid errors**
- **HW**: review **WLS**
- review **GLS**

GLS for Long. data when Σ is known

$$y_i \sim N(\mu_i, \Sigma) \quad i = 1, 2, \ldots, N$$

where $\mu_i = X_i\beta$ ($n_i \times p$ and $p \times 1$)

$$f(y_i; \mu_i, \Sigma) = \left(\frac{1}{2\pi}\right)^{n_i/2} \frac{1}{|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(y_i - \mu_i)^T \Sigma^{-1} (y_i - \mu_i)\right\}$$

when Σ is known, we obtain the MLE for β (from $\mu_i = X_i\beta$) via

$$\max_\beta L(\beta) = \prod_{i=1}^{N} \left(\frac{1}{2\pi}\right)^{n_i/2} |\Sigma|^{-1/2} \exp\left\{-\frac{1}{2}(y_i - \mu_i)^T \Sigma^{-1} (y_i - \mu_i)\right\}$$

$$= \left(\frac{1}{2\pi}\right)^{\sum \mathbf{n_i}/2} |\Sigma|^{-N/2} \exp\left\{-\frac{1}{2}\sum_{i=1}^{N}(y_i-\mu_i)^T\Sigma^{-1}(y_i-\mu_i)\right\}$$

> [!TIP] The writer is simplifying the likelihood function by grouping the constants and exponents. Recall that for independent observations, the likelihood $L(\beta)$ is the product of the individual probability density functions. The rule $\prod e^{a_i} = e^{\sum a_i}$ allows the final consolidation of terms into the exponential.

Let $K = \sum_{i=1}^{N} n_i$.

$$L(\beta) \propto \sum_{i=1}^{N}(y_i - X_i\beta)^T\Sigma^{-1}(y_i - X_i\beta)$$

$$= \sum_{i=1}^{N}\{y_i^T\Sigma^{-1}y_i - y_i^T\Sigma^{-1}X_i\beta - \beta^T X_i^T\Sigma^{-1}y_i + \beta^T X_i^T\Sigma^{-1}X_i\beta\}$$

$$= \sum_{i=1}^{N} y_i^T\Sigma^{-1}y_i - 2\sum_{i=1}^{N}\beta^T X_i^T\Sigma^{-1}y_i + \sum_{i=1}^{N}\beta^T X_i^T\Sigma^{-1}X_i\beta$$

$$\hat{\beta}_{MLE} = \text{argmax}_\beta \left\{\sum \beta^T X_i^T\Sigma^{-1}X_i\beta - 2\sum \beta^T X_i^T\Sigma^{-1}y_i\right\}$$

$$\begin{aligned}\frac{\partial}{\partial\beta} : \sum_{i=1}^{N}&\{\beta^T X_i^T\Sigma^{-1}X_i + [X_i^T\Sigma^{-1}X_i\beta]^T\} \\ &- 2\sum_{i=1}^{N}[X_i^T\Sigma^{-1}y_i]^T \\ &= \sum_{i=1}^{N}\{2\beta^T X_i^T\Sigma^{-1}X_i - 2y_i^T\Sigma^{-1}X_i\} \\ &= 2\left[\beta^T\sum X_i^T\Sigma^{-1}X_i - \sum y_i^T\Sigma^{-1}X_i\right]\end{aligned}$$

$$\Rightarrow \beta^T = \left[\sum y_i^T\Sigma^{-1}X_i\right]\left[\sum X_i^T\Sigma^{-1}X_i\right]^{-1}$$

or $\beta = \left[\sum_{i=1}^{N} X_i^T\Sigma^{-1}X_i\right]^{-1}\left(\sum_{i=1}^{N} X_i^T\Sigma^{-1}y_i\right)$

If we stack $\{Y_i\}$, we get

$$\underset{K\times 1}{y} = \underset{K\times p}{X}\underset{p\times 1}{\beta} + \underset{K\times 1}{e}$$

> [!TIP] Stacking observations $\{Y_i\}$ into a vector y is the standard matrix formulation of the **General Linear Model**. This representation transforms K independent regression equations into a single compact equation.

with $e \sim N_K\left(0, \begin{pmatrix} \Sigma & & 0 \\ & \ddots & \\ 0 & & \Sigma \end{pmatrix}\right)$.

> [!TIP] The block diagonal covariance matrix implies that the error vectors for different observations are uncorrelated (and independent, given the multivariate normal assumption).

Include a review of differentiation wrt a vector $\beta_{p\times 1}$.

If $g(\beta) \in \mathbb{R}$, i.e., $g : \mathbb{R}^p \to \mathbb{R}$, then

$$\underset{p\times 1}{\nabla g} = \begin{pmatrix} \frac{\partial g}{\partial \beta_1} \\ \vdots \\ \frac{\partial g}{\partial \beta_p} \end{pmatrix}$$

$$\nabla^2 g = \begin{pmatrix} \frac{\partial^2 g}{\partial \beta_1 \partial \beta_1} & \frac{\partial^2 g}{\partial \beta_1 \partial \beta_2} & \cdots & \frac{\partial^2 g}{\partial \beta_1 \partial \beta_p} \\ & \ddots & & \end{pmatrix}$$

e.g. $g(\beta) = a^T\beta = \sum_{j=1}^p a_j\beta_j$ ($1 \times p$ and $p \times 1$)

$$\frac{\partial g}{\partial \beta_j} = a_j \quad \text{and} \quad \nabla_\beta g = a$$

Consider $g(\beta) = A\beta \in \mathbb{R}^m$ ($m \times p$)

> [!TIP] Here A is an $m \times p$ matrix and β is a $p \times 1$ vector, resulting in a vector in $\mathbb{R}^m$. The expression $g(\beta)_j$ represents the j-th component of this resulting vector.

$$g(\beta)_j = \sum_{k=1}^p a_{jk}\beta_k$$

so $\frac{\partial}{\partial \beta_s} g(\beta)_j = a_{js}$

> [!TIP] To calculate this partial derivative, recall that $\frac{\partial}{\partial \beta_s}\sum_{k=1}^p a_{jk}\beta_k = \sum_{k=1}^p a_{jk}\frac{\partial \beta_k}{\partial \beta_s}$. The derivative $\frac{\partial \beta_k}{\partial \beta_s}$ is 1 when $k = s$ and 0 otherwise (Kronecker delta δ_{ks}), so the sum collapses to a_{js}.

2.2 GLS estimator vs. MLE of β

How to estimate $\Sigma = \Sigma(\theta)$ if using GLS for β?

2.3 MCAR vs. MAR

Assign a problem where data are dropped at random (MCAR) & then systematically related to observed $\{y\}$ - MAR. Compare resulting estimates of β, θ.

> [!TIP] **MCAR (Missing Completely At Random)** means the probability of data being missing is independent of both observed and unobserved data. **MAR (Missing At Random)** means the probability of missing data is related to the observed data $\{y\}$ but not the unobserved data. This distinction is critical because MAR data can often be handled using likelihood-based methods, whereas MCAR is a stronger, more restrictive assumption.

The text motivates GLS using ML. This does not mirror the development from LS. Reconsider with LS motivation leading to GLS.

By GLS, does the text **ALWAYS** mean that Σ is known?

2.4 REML vs. ML

Show REML vs. ML est. effect on θ & that this effect reduces as $N \uparrow$ ($N =$ #subjects).

Derive the REML loglikelihood given in the text (4.8).

The REML likelihood is the likelihood of $r = y - \hat{y}$ after β has been estimated by OLS, i.e. $\hat{y} = \hat{y}(\hat{\beta}_{OLS}) = X\hat{\beta}_{OLS}$.

Let's consider subject i.

$$y_i \sim N(x_i\beta, \Sigma)$$

?

$$\hat{\beta}_{OLS} = \left(\sum_{i=1}^{N} x_i^T x_i\right)^{-1} \left(\sum_{i=1}^{N} (x_i^T x_i)^{-1} x_i^T y_i\right)$$

$$= (X^T X)^{-1} X^T y$$

$$E(\hat{\beta}_{OLS}) = (X^TX)^{-1}X^TX\beta = \beta$$

?

$$Cov(\hat{\beta}_{OLS}) = \left((X^TX)^{-1}X^T\right)\Sigma\left((X^TX)^{-1}X^T\right)^T$$

The dimensions are indicated below:

$$p \times p \quad p \times n \qquad n \times n \qquad n \times p \qquad n \times p$$

(Note: The last term in the image appears as $(n \times p)^T = p \times n$.)

$$= (X^TX)^{-1}X^T\Sigma X(X^TX)^{-1}$$

$$E(r) = E(y - \hat{y}) = X\beta - X\beta = 0$$

?

$$\begin{aligned} Cov(r) &= Cov(y - \hat{y}) \\ &= Cov(X[\beta - \hat{\beta}_{OLS}]) = Cov(-X\hat{\beta}_{OLS}) \\ &= Cov(X\hat{\beta}_{OLS}) = XCov(\hat{\beta}_{OLS})X^T \\ &= X\{(X^TX)^{-1}X^T\Sigma X(X^TX)^{-1}\}X^T \quad \checkmark \end{aligned}$$

> [!CAUTION] The derivation of $Cov(r)$ is incorrect. The substitution $Cov(y - \hat{y}) = Cov(X(\beta - \hat{\beta}_{OLS}))$ assumes $y = X\beta$ (i.e., no noise ϵ), ignoring the error term. The true variance of the residuals $r = (I - P)y$ is $(I - P)\Sigma(I - P)^T$, not $XCov(\hat{\beta})X^T$.

$$= \Sigma$$

> [!CAUTION] The conclusion $Cov(r) = \Sigma$ is incorrect; the variance of the residuals is reduced by the projection onto the column space of X.

Implies

$$L(\Sigma) = \left(\frac{1}{2\pi}\right)^{\frac{K}{2}} |\Sigma|^{-\frac{N}{2}} \exp\left\{-\frac{1}{2}\sum_{i=1}^{N} r_i^T\Sigma^{-1}r_i\right\}$$

$$\ell(\Sigma) = -\frac{N}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{N} r_i^T\Sigma^{-1}r_i$$

I don't have the 3rd term here

> [!TIP] The missing "3rd term" in the log-likelihood $\ell(\Sigma)$ is the constant $-\frac{NK}{2}\log(2\pi)$ (or $-\frac{K}{2}\log(2\pi)$ depending on normalization convention) derived from the Gaussian prefactor $\left(\frac{1}{2\pi}\right)^{\frac{K}{2}}$.

Extend Chp 5 to more than one grouping factor & having covariates to adjust for.

$$\hat{\beta}_{OLS}, r_i = y_i - \hat{y}_i(\hat{\beta}_{OLS}) = y_i - x_i\hat{\beta}_{OLS}$$

$$E(r_i) = E(y_i) - x_i E(\hat{\beta}_{OLS})$$

$$E(\hat{\beta}_{OLS}) = E\left[\left\{\sum_{i=1}^{N} x_i^T x_i\right\}^{-1} \sum_{i=1}^{N} x_i^T y_i\right]$$

$$= \left\{\sum_{i=1}^{N} x_i^T x_i\right\}^{-1} \sum_{i=1}^{N} x_i^T x_i \beta = \beta$$

$$\therefore E(r_i) = E(y_i) - x_i\beta = 0$$

$$Cov(r_i) = Cov(y_i - x_i\hat{\beta}_{OLS})$$

$$= E[(y_i - x_i\hat{\beta}_{OLS})(y_i - x_i\hat{\beta}_{OLS})^T]$$

$$= E[(y_i - x_i\beta + x_i\beta - x_i\hat{\beta})(y_i - x_i\beta + x_i\beta - x_i\hat{\beta})^T]$$

$$= E[((y_i - x_i\beta) + x_i(\beta - \hat{\beta}))((y_i - x_i\beta) + x_i(\beta - \hat{\beta}))^T]$$

$$\begin{aligned} = E[&(y_i - x_i\beta)(y_i - x_i\beta)^T + (y_i - x_i\beta)[x_i(\beta - \hat{\beta})]^T \\ &+ x_i(\beta - \hat{\beta})(y_i - x_i\beta)^T + x_i(\beta - \hat{\beta})(\beta - \hat{\beta})^T x_i^T] \end{aligned}$$

$$= \Sigma_i + E[(y_i - x_i\beta)(\beta - \hat{\beta})^T x_i^T] + E[x_i(\beta - \hat{\beta})(y_i - x_i\beta)^T] + x_i Cov(\hat{\beta}) x_i^T$$

$$A = E[(y_i - x_i\beta)(\beta - \hat{\beta})^T x_i^T - x_i\beta(\beta - \hat{\beta})^T x_i^T]$$

$$= x_i \beta \beta^T x_i^T - E[y_i \widehat{\beta}^T x_i^T] - x_i \beta \beta^T x_i^T + x_i \beta \beta^T x_i^T$$

$$= x_i \beta \beta^T x_i^T - E[y_i \widehat{\beta}^T] x_i^T$$

$$E[y_i \widehat{\beta}^T] = E\left[y_i \left(\sum_{j=1}^{N} x_j^T x_j \right)^{-1} \sum_{j=1}^{N} x_j^T y_j^T \right]$$

$$E\left\{ y_i \left[\sum_{j=1}^{N} x_j^T x_j \right]^{-1} \left\{ \sum_{j=1}^{N} x_j^T x_j^T \right\} \right\}$$

$$n \times p \quad 1 \times p \quad p \times p$$

$$= E\{Y_i Y_i^T X_i \{\sum_j X_j^T X_j\}^{-1}\}$$

$$= E\{Y_i Y_i^T\} X_i \{\sum_j X_j^T X_j\}^{-1}$$

[!TIP] The writer is manipulating the expectation of the OLS estimator $\hat{\beta} = (\sum X_j^T X_j)^{-1} \sum X_i^T Y_i$. Factoring out the deterministic parts (the X matrices) from the expectation is standard linearity of expectation.

$$\text{Cov}(Y_i) = E[(Y_i - \mu_i)(Y_i - \mu_i)^T]$$

$$= E[Y_i Y_i^T - Y_i \mu_i^T - \mu_i Y_i^T + \mu_i \mu_i^T]$$

$$= E[Y_i Y_i^T] - 2\mu_i \mu_i^T + \mu_i \mu_i^T$$

$$= E[Y_i Y_i^T] - \mu_i \mu_i^T$$

$$\Rightarrow E[Y_i Y_i^T] = \Sigma_i + \mu_i \mu_i^T$$

[!TIP] This is the computational formula for variance: $\text{Var}(X) = E[X^2] - (E[X])^2$. Here it is applied to the vector case: $\text{Cov}(Y) = E[YY^T] - E[Y]E[Y]^T$.

$$E(Y_i\hat{\beta}^T) = [\Sigma_i + \mu_i\mu_i^T]X_i\{\sum_j X_j^T X_j\}^{-1}$$

$$A = X_i\beta\beta^T X_i^T - [\Sigma_i + \mu_i\mu_i^T]X_i\{\sum_j X_j^T X_j\}^{-1}X_i^T$$

$$\mathrm{Cov}(r_i) = \Sigma_i + A + A^T + X_i\mathrm{Cov}(\hat{\beta})X_i^T$$

$$\mathrm{Cov}(\hat{\beta}_{\mathrm{ols}}) = \mathrm{Cov}(M^{-1}\sum_{j=1}^N X_j^T Y_j)$$

$$= M^{-1}\mathrm{Cov}(\sum_{j=1}^N X_j^T Y_j)M^{-T}$$

$$\mathrm{Cov}(\sum_{i=1}^N X_i^T Y_i) = E[(\sum X_i^T Y_i - \sum X_i^T X_i\beta)(\sum X_i^T Y_i - \sum X_i^T X_i\beta)^T]$$

$$\begin{aligned}= E[&\sum X_i^T Y_i(\sum X_i^T Y_i)^T - \sum X_i^T Y_i(\sum X_i^T X_i\beta)^T \\ &-(\sum X_i^T X_i\beta)(\sum X_i^T Y_i)^T \\ &+(\sum X_i^T X_i\beta)(\sum X_i^T X_i\beta)^T]\end{aligned}$$

$$= E[\sum_{i=1}^N X_i^T Y_i Y_i^T X_i + N X_i^T Y_i(\sum_{j\neq i} X_j^T X_j)^T]$$

$$\begin{aligned}&-\sum x_i^T x_i\beta\beta^T(\sum x_i^T x_i)^T \\ &-\sum x_i^T x_i\beta\beta^T(\sum x_i^T x_i)^T \\ &+\sum x_i^T x_i\beta\beta^T(\sum x_i^T x_i)^T \\ =&\sum_i x_i^T x_i\beta\beta^T x_i^T x_i + N x_i^T x_i\beta(\sum_{j\neq i} x_j^T x_j\beta)^T \\ &-(\sum x_i^T x_i)\beta\beta^T(\sum x_i^T x_i)^T\end{aligned}$$

Since $E(r_i) = 0$, we have $Cov(r_i) = E[r_i r_i^T]$.

If we stack all y_i, all x_i, we have

$$y = X\beta + e$$

$N \times 1$ $\quad$ $N \times p$ $\quad$ $p \times 1$ $\quad$ $N \times 1$

and it is clear that $\widehat{\beta} = (X^T X)^{-1} X^T y$.

Also $\widehat{y} = X\widehat{\beta} = X(X^T X)^{-1} X^T y = Hy$.

$r = y - \widehat{y} = (I - H)y$.

$$Cov(r) = E[(I-H)yy^T(I-H)^T]$$

$$= E[(I-H)yy^T(I-H)]$$

$$= (I-H)(\Sigma^* + \mu\mu^T)(I-H) = (I-H)\Sigma^*(I-H)$$

(let $K = Nn$)

As we see that $r \sim N(0, \Sigma^r)$. So the likelihood for r is

$$\left(\frac{1}{2\pi}\right)^{K/2} |\Sigma^r|^{-1/2} \exp\left\{-\frac{1}{2} r^T (\Sigma^r)^{-1} r\right\}$$

$$1 \times K \quad K \times K \quad K \times 1$$

Loglikelihood is

$$-\frac{1}{2}\log|\Sigma^r| - \frac{1}{2} r^T (\Sigma^r)^{-1} r$$

$$= -\frac{1}{2}\log|(I-H)\Sigma^*(I-H) + (I-H)\mu\mu^T(I-H)| - \frac{1}{2} r^T (\Sigma^r)^{-1} r$$

$$= -\frac{1}{2}\log\bigg| I\Sigma^* I - H\Sigma^* I + H\Sigma^* H$$

$$- I\Sigma^* H\bigg|$$

$$- \frac{1}{2} r^T (\Sigma^r)^{-1} r$$

$$= -\frac{1}{2}\log\bigg|\Sigma^* - H\Sigma^* - \Sigma^* H + H\Sigma^* H\bigg|$$

$$- \frac{1}{2} r^T (\Sigma^r)^{-1} r$$

$$\Sigma^r = (I-H)\Sigma^*(I-H).$$

2.5 OLS

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad \hat{Y} = X\hat{\beta} = HY$$

$$r = Y - \hat{Y} = (I - H)Y$$

$$E(r) = 0$$

$$\begin{aligned}
\mathrm{Var}(r) &= \mathrm{Var}((I-H)Y) \\
&= (I-H)(\sigma^2 I)(I-H)^T \\
&= (I-H)\sigma^2
\end{aligned}$$

$$H = X(X^T X)^{-1} X^T$$

$$I - H = I - X(X^T X)^{-1} X^T$$

2.6 Likelihood based on Y:

$$\left(\frac{1}{2\pi}\right)^{n/2} |\sigma^2 I|^{-1/2} \exp\left\{-\frac{1}{2}(Y - X\beta)^T (\sigma^2 I)^{-1} (Y - X\beta)\right\}$$

Log-likelihood is:

$$-\frac{n}{2}\log(2\pi) - \frac{1}{2}\log|\sigma^2 I| - \frac{1}{2\sigma^2}(Y - X\beta)^T (Y - X\beta)$$

2.7 Likelihood based on r

$$\left(\frac{1}{2\pi}\right)^{n/2} |(I-H)\sigma^2|^{-1/2} \exp\left\{-\frac{1}{2} r^T [(I-H)\sigma^2]^{-1} r\right\}$$

$$= (\sigma^{2n})^{-1/2} |I - H|^{1/2} \exp\left\{-\frac{1}{2\sigma^2} r^T (I-H)^{-1} r\right\}$$

> [!CAUTION] The projection matrix $(I - H)$ is not full rank (it has rank $n - p$), so it is singular and not invertible. The inverse $(I - H)^{-1}$ does not exist.

$$= (\sigma^2)^{-n/2}|I-H|^{1/2}\exp\left\{-\frac{1}{2\sigma^2}r^T(I-H)^{-1}r\right\}$$

Let $\theta = \sigma^2$:

$$L(\theta) = \theta^{-n/2}|I-H|^{1/2}\exp\left\{-\frac{1}{2\theta}r^T(I-H)^{-1}r\right\}$$

> [!TIP] The residual vector r is orthogonal to the subspace spanned by X, meaning r lies in an $(n-p)$-dimensional subspace. Because of this linear dependence, you cannot define a standard likelihood for r using the inverse of $(I-H)$. You would typically project r into its own orthonormal basis to remove the singularity.

$$\begin{aligned}
l(\theta) &= -\frac{n}{2}\log\theta - \frac{1}{2\theta}r^T(I-H)^{-1}r \\
\frac{\partial}{\partial\theta}l(\theta) &= -\frac{n}{2\theta} + \frac{1}{2\theta^2}r^T(I-H)^{-1}r = 0 \\
-n + \frac{1}{\theta}r^T(I-H)^{-1}r = 0 \\
\theta &= \frac{r^T(I-H)^{-1}r}{n}
\end{aligned}$$

> [!TIP] This is a **Maximum Likelihood Estimation (MLE)** derivation for the variance parameter θ. The estimator is obtained by setting the derivative of the log-likelihood function to zero and solving for θ.

$$\begin{aligned}
\mathrm{Cov}(\hat{\beta}_{\mathrm{ols}}) &= \mathrm{Cov}((X^TX)^{-1}X^TY) \\
&= (X^TX)^{-1}X^T\mathrm{Cov}(Y)X(X^TX)^{-1} \\
&= (X^TX)^{-1}X^T\Sigma X(X^TX)^{-1}
\end{aligned}$$

> [!TIP] The covariance of a linear transformation of a random variable Y is given by $\mathrm{Cov}(AY) = A\Sigma A^T$. Here, $A = (X^TX)^{-1}X^T$. Since $(X^TX)^{-1}$ is symmetric, its transpose is itself, and the transpose of X^T is X, yielding $A^T = X(X^TX)^{-1}$.

$$\begin{aligned}
&\text{If } \Sigma = \sigma^2 I, \text{ then} \\
\mathrm{Cov}(\hat{\beta}) &= (X^TX)^{-1}X^TX(X^TX)^{-1} \\
&= (X^TX)^{-1}
\end{aligned}$$

> [!CAUTION] Algebra error: The substitution $\Sigma = \sigma^2 I$ was made, but the scalar σ^2 was dropped from the expression. The result should be $\sigma^2(X^TX)^{-1}$, not $(X^TX)^{-1}$.

3 REML estimation

Consider first simple linear regression.

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

$$e_i \sim N(0, \sigma^2) \text{ i.i.d.}$$

The likelihood is

$$\begin{aligned}
L(\beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\sigma^2}\right)^{1/2} \exp\left\{\frac{-1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2\right\} \\
&= \left(\frac{1}{2\pi}\right)^{N/2} \left(\frac{1}{\sigma^2}\right)^{N/2} \exp\left\{\frac{-1}{2\sigma^2}\sum_{i=1}^{N}(y_i - \beta_0 - \beta_1 x_i)^2\right\}
\end{aligned}$$

So

$$l(\beta_0, \beta_1, \sigma^2) = -\frac{N}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - (\beta_0 + \beta_1 x_i))^2$$

> [!TIP] The log-likelihood l here effectively drops the constant term $-\frac{N}{2}\log(2\pi)$. This is standard in maximum likelihood estimation (MLE) because constants do not affect the value of parameters that maximize the likelihood.

and

$$\begin{aligned}
\frac{\partial l}{\partial \sigma^2} &= -\frac{N}{2}\frac{1}{\sigma^2} - \frac{1}{2}(-1)(\sigma^2)^{-2}\sum_{i=1}^{N}(y_i - (\beta_0 + \beta_1 x_i))^2 \\
\Rightarrow N &= \frac{1}{\sigma^2} RSS(\beta_0, \beta_1)
\end{aligned}$$

and

$$\hat{\sigma}^2_{\text{MLE}} = \frac{RSS(\beta_0, \beta_1)}{N} \quad \textbf{BIASED}$$

> [!TIP] This estimator is biased because it uses N (total observations) in the denominator rather than $N-p$ (degrees of freedom, where $p=2$ for simple linear regression). The bias arises because MLE does not account for the degrees of freedom lost when estimating the regression coefficients β_0 and β_1. REML (Restricted Maximum Likelihood) estimation is often used to correct this by effectively ignoring the fixed effects during variance estimation.

REML partitions the likelihood to separate the contributions to estimation of σ^2 and the contributions to the estimation of β_0, β_1.

To see that this is possible, note:

$$\begin{aligned}
&\sum_{i=1}^{N}(y_i - (\beta_0 + \beta_1 x_i))^2 \\
&= \sum_{i=1}^{N}(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) + (\hat{\beta}_0 + \hat{\beta}_1 x_i) - (\beta_0 + \beta_1 x_i))^2 \\
&= \sum_{i=1}^{N}(y_i - \hat{y}_i + \hat{y}_i - (\beta_0 + \beta_1 x_i))^2
\end{aligned}$$

where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ using the MLEs for β_0, β_1

$$\begin{aligned}
&= \sum_{i=1}^{N}\{(y_i - \hat{y}_i)^2 + (\hat{y}_i - (\beta_0 + \beta_1 x_i))^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - (\beta_0 + \beta_1 x_i))\} \\
&= \sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{N}(\hat{y}_i - (\beta_0 + \beta_1 x_i))^2
\end{aligned}$$

> [!TIP] The cross-term $2\sum_{i=1}^{N}(y_i - \hat{y}_i)(\hat{y}_i - (\beta_0 + \beta_1 x_i))$ vanishes because of the properties of OLS residuals: they are orthogonal to the fitted values (and the regressors), meaning the sum of their product is zero.

Consequently, the loglikelihood can be written as

$$l(\beta_0, \beta_1, \sigma^2) = -\frac{N}{2}\log\sigma^2 + \frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(\hat{y}_i - (\beta_0 + \beta_1 x_i))^2$$

Now consider $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Since $\hat{\beta}_0$ and $\hat{\beta}_1$ are normally distributed, so is $\hat{y}_i$. Also, $\mathrm{Cov}\left(\begin{pmatrix}\hat{\beta}_0 \\ \hat{\beta}_1\end{pmatrix}\right) = \sigma^2(X^T X)^{-1}$.

Thus $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \sigma^2\left[(X^T X)^{-1}\right]_{12}$.

$$(X^TX) = \begin{pmatrix} N & \sum_{i=1}^N x_i \\ \sum_{i=1}^N x_i & \sum_{i=1}^N x_i^2 \end{pmatrix}, \text{ so}$$

$$(X^TX)^{-1} = \frac{1}{N\sum x_i^2 - (\sum x_i)^2} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & N \end{pmatrix}$$

> [!TIP] This is the formula for the inverse of a 2×2 matrix: for a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

$$\Rightarrow \text{Note: } N(\sum x_i^2) - (\sum x_i)^2 = N \cdot SXX$$

$$\begin{aligned} SXX &= \sum (x_i - \bar{x})^2 = \sum (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \sum x_i^2 - 2\bar{x}\sum x_i + N\bar{x}^2 \\ &= \sum x_i^2 - 2\bar{x}(N\bar{x}) + N\bar{x}^2 \\ &= \sum x_i^2 - 2N\bar{x}^2 + N\bar{x}^2 \\ &= \sum x_i^2 - N\bar{x}^2 \\ &= \sum x_i^2 - N\left(\frac{1}{N}\right)^2 (\sum x_i)^2 \\ &= \sum x_i^2 - \frac{1}{N}(\sum x_i)^2 \end{aligned}$$

So

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \frac{1}{N \cdot SXX}\sum X_i^2 \cdot \sigma^2 \\ \text{Var}(\hat{\beta}_1) &= \frac{1}{N \cdot SXX} \cdot N = \frac{1}{SXX} \cdot \sigma^2 \end{aligned}$$

> [!CAUTION] The variance formula is missing the σ^2 term in the first two parts of the equation; it only appears after the final equals sign.

$$\begin{aligned} \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \frac{1}{N \cdot SXX}(-N\bar{X}) \cdot \sigma^2 \\ &= -\sigma^2 \frac{\bar{X}}{SXX} \end{aligned}$$

Thus we see that $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ has mean $\beta_0 + \beta_1 x_i$ and variance

$$\frac{\sigma^2}{SXX}\left\{\frac{\sum x_i^2}{N}+x_i^2-x_i\cdot 2\bar{X}\right\}$$

> [!TIP] This variance expression can be simplified to the standard prediction interval formula: $\text{Var}(\hat{y}_i)=\sigma^2\left(\frac{1}{N}+\frac{(x_i-\bar{x})^2}{SXX}\right)$. The written sum simplifies to the squared deviation from the mean.

In matrix form,

$$y = X\beta + e, \quad e\sim N(0,\sigma^2 I_n)$$

$$L(\beta,\sigma^2)=\left(\frac{1}{2\pi}\right)^{N/2}\left(\frac{1}{\sigma^2}\right)^{N/2}\exp\left\{\frac{-1}{2\sigma^2}(y-X\beta)^T(y-X\beta)\right\}$$

$$\ell(\beta,\sigma^2)=-\frac{N}{2}\log(2\pi)-\frac{N}{2}\log(\sigma^2)-\frac{1}{2\sigma^2}(y-X\beta)^T(y-X\beta)$$

Writing $(y-X\hat{\beta}+X\hat{\beta}-X\beta)^T(y-X\hat{\beta}+X\hat{\beta}-X\beta)$

$$\begin{aligned}
&=(y-X\hat{\beta})^T(y-X\hat{\beta})+(X\hat{\beta}-X\beta)^T(X\hat{\beta}-X\beta)\\
&\quad+(y-X\hat{\beta})^T(X\hat{\beta}-X\beta)+(X\hat{\beta}-X\beta)^T(y-X\hat{\beta})\\
&=(y-X\hat{\beta})^T(y-X\hat{\beta})+(\hat{\beta}-\beta)^TX^TX(\hat{\beta}-\beta)
\end{aligned}$$

> [!TIP] The cross terms $(y-X\hat{\beta})^T(X\hat{\beta}-X\beta)$ and $(X\hat{\beta}-X\beta)^T(y-X\hat{\beta})$ vanish because of the **Normal Equations**, which state that $X^T(y-X\hat{\beta})=0$. This implies that the residual vector is orthogonal to the column space of X.

We see

$$\begin{aligned}
\ell(\beta,\sigma^2)=&-\frac{N}{2}\log(2\pi)-\frac{N}{2}\log(\sigma^2)\\
&-\frac{1}{2\sigma^2}(y-X\hat{\beta})^T(y-X\hat{\beta})\\
&-\frac{1}{2\sigma^2}(\hat{\beta}-\beta)^TX^TX(\hat{\beta}-\beta)
\end{aligned}$$

Now $r=y-X\hat{\beta}$ has a normal distribution (though degenerate!).

$$E(y-X\hat{\beta})=X\beta-X\hat{\beta}=0$$

> [!TIP] The residual vector $r = y - X\hat{\beta}$ is "degenerate" because it lies in the subspace of $\mathbb{R}^n$ that is orthogonal to the column space of X. While it follows a normal distribution, its variance is constrained to that $(n-p)$-dimensional subspace, not the full $\mathbb{R}^n$.

(Simple Linear Regression)

$$\begin{aligned}\mathrm{Var}(Y - X\hat{\beta}) &= \mathrm{Var}((I-H)Y) \\ &= (I-H)\sigma^2 I (I-H)^T \\ &= \sigma^2 (I-H)\end{aligned}$$

Show the likelihood derived from r is

$$\left(\frac{1}{2\pi}\right)^{\frac{N-1}{2}} |\sigma^2(I-H)|^{-1/2} \exp\left\{-\frac{1}{2\sigma^2} r^T r\right\}$$

$$l(\sigma^2) = -\frac{N}{2}\log|\sigma^2(I-H)| - \frac{1}{2\sigma^2} r^T r$$

$$= -\frac{1}{2}\log\{|\sigma^2(I-H)|\} - \frac{1}{2\sigma^2} r^T r$$

$$= -\frac{1}{2}\{(N-1)\log\sigma^2 + \log|I-H|\} - \frac{1}{2\sigma^2} r^T r$$

den full likelihood is

$$l(\beta, \sigma^2) = -\frac{N}{2}\log(2\pi) - \frac{N}{2}\log\sigma^2 - \frac{1}{2\sigma^2} r^T r - \frac{1}{2\sigma^2}(\hat{\beta}-\beta)^T X^T X (\hat{\beta}-\beta)$$

$$\begin{aligned}&= -\frac{N-1}{2}\log(2\pi) - \frac{1}{2}\log(2\pi) \\ &\quad - \frac{N-1}{2}\log\sigma^2 - \frac{1}{2}\log\sigma^2 \\ &\quad - \frac{1}{2\sigma^2} r^T r - \frac{1}{2}\log|\sigma^2(I-H)| \\ &\quad - \frac{1}{2\sigma^2}(\hat{\beta}-\beta)^T X^T X (\hat{\beta}-\beta)\end{aligned}$$

> [!TIP] The simplification $r^T(I-H)r = r^T r$ used in the derivation relies on the property that the residual vector r is orthogonal to the column space of X, so $Hr = 0$. Since $r = (I-H)Y$, we have $r^T(I-H)r = r^T r - r^T H r = r^T r - 0 = r^T r$.

$$l(\beta, \sigma^2) = l_r(\sigma^2) - \frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}(\hat{\beta} - \beta)^T X^T X(\hat{\beta} - \beta)$$

Maximizing $l_r(\sigma^2)$ gives

$$\begin{aligned}
\frac{\partial l_r}{\partial \sigma^2} &= -\frac{1}{2}(N-1)\log\sigma^2 - \frac{1}{2\sigma^2}r^T r - \frac{1}{2}\log|\sigma^2(I-H)| \\
&= -\frac{1}{2}(N-1)\log\sigma^2 - \frac{1}{2\sigma^2}r^T r - \frac{1}{2}(N-1)\log\sigma^2 - \frac{1}{2}\log|\sigma^2(I-H)|
\end{aligned}$$

> [!CAUTION] The expression on the right-hand side of the derivative equation is the log-likelihood function itself, not its derivative. The derivative with respect to σ^2 should be a function of σ^2 (typically involving $1/\sigma^2$).

Consider multiple linear regression (OLS):

$$y = X\beta + e, \quad e \sim N(0, \sigma^2 I_N)$$

The likelihood function is

$$\begin{aligned}
L(\beta, \sigma^2) &= \left(\frac{1}{2\pi}\right)^{N/2} |\sigma^2 I_N|^{-1/2} \exp\left\{-\frac{1}{2}(y - X\beta)^T(\sigma^2 I_N)^{-1}(y - X\beta)\right\} \\
&= \left(\frac{1}{2\pi}\right)^{N/2} (\sigma^2)^{-N/2} \exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^T(y - X\beta)\right\}
\end{aligned}$$

$$l(\beta, \sigma^2) = -\frac{N}{2}\log(2\pi) - \frac{N}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)^T(y - X\beta)$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{N}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{1}{(\sigma^2)^2}(y - X\beta)^T(y - X\beta)$$

$$\Rightarrow -N + \frac{1}{\sigma^2}\mathrm{RSS}(\beta) = 0$$

$$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{RSS}(\hat{\beta}_{\mathrm{MLE}})}{N} \quad \textbf{BIASED!}$$

NOTE:

$$
\begin{aligned}
(y - X\beta)^T(y - X\beta) &= (y - X\hat{\beta} + X\hat{\beta} - X\beta)^T(y - X\hat{\beta} + X\hat{\beta} - X\beta) \\
&= (y - X\hat{\beta})^T(y - X\hat{\beta}) \\
&\quad + 2(y - X\hat{\beta})^T X(\hat{\beta} - \beta) \\
&\quad + (\hat{\beta} - \beta)^T X^T X(\hat{\beta} - \beta)
\end{aligned}
$$

> [!TIP] The expansion of the sum of squared errors decomposes the total error into the residual sum of squares and the parameter estimation error. In the cross-term $2(y - X\hat{\beta})^T X(\hat{\beta} - \beta)$, the component $(y - X\hat{\beta})^T X$ is the zero vector, which is a key property of the OLS normal equations: the residuals are orthogonal to the columns of the design matrix X.

Middle term is

$$
2(X^T(Y - X\hat{\beta}))^T(\hat{\beta} - \beta) = 2(\underbrace{X^T Y - X^T X\hat{\beta}}_{0})^T(\hat{\beta} - \beta) = 0
$$

> [!TIP] This term is zero because $\hat{\beta}$ satisfies the normal equations $X^T X\hat{\beta} = X^T Y$, so $X^T Y - X^T X\hat{\beta} = 0$. This is the first-order condition for the ordinary least squares estimator.

So the loglikelihood is

$$
\begin{aligned}
&-\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\sigma^2 \\
&-\frac{1}{2\sigma^2}(Y - X\hat{\beta})^T(Y - X\hat{\beta}) \\
&-\frac{1}{2\sigma^2}(\hat{\beta} - \beta)(X^T X)(\hat{\beta} - \beta)
\end{aligned}
$$

> [!CAUTION] Missing transpose: the term $(\hat{\beta} - \beta)(X^T X)(\hat{\beta} - \beta)$ should be written as $(\hat{\beta} - \beta)^T(X^T X)(\hat{\beta} - \beta)$ to correctly denote the inner product/quadratic form.

note $(Y - X\hat{\beta})^T(Y - X\hat{\beta})$

$$
= ((I - H)Y)^T(I - H)Y
$$

$$
= Y^T(I - H)Y
$$

$$
l(\beta, \sigma^2) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}Y^T(I - H)Y - \frac{1}{2\sigma^2}(\hat{\beta} - \beta)(X^T X)(\hat{\beta} - \beta)
$$

> [!CAUTION] Again, the term $(\hat{\beta}-\beta)(X^TX)(\hat{\beta}-\beta)$ is missing the transpose on the first factor.

$$
\begin{aligned}
&= -\frac{P}{2}\ln(2\pi) - \frac{(N-P)}{2}\ln(2\pi) \\
&\quad - \frac{P}{2}\ln\sigma^2 - \frac{(N-P)}{2}\ln\sigma^2 \\
&\quad - \frac{1}{2\sigma^2}(\hat{\beta}-\beta)(X^TX)(\hat{\beta}-\beta) - \frac{1}{2\sigma^2}Y^T(I-H)Y
\end{aligned}
$$

> [!CAUTION] The term $(\hat{\beta}-\beta)(X^TX)(\hat{\beta}-\beta)$ is missing the transpose on the first factor.

The red terms are the residual likelihood. $l_r(\sigma^2)$

$$
\begin{aligned}
\frac{\partial}{\partial\sigma^2}l_r(\sigma^2) &= -\frac{(N-p)}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{1}{(\sigma^2)^2}y^T(I-H)y \\
&\Rightarrow -(N-p) + \frac{1}{\sigma^2}y^T(I-H)y \\
\hat{\sigma}^2 &= \frac{y^T(I-H)y}{N-p} \quad \checkmark
\end{aligned}
$$

> [!TIP] The second step involves setting the derivative to zero to find the maximum likelihood estimator ($\hat{\sigma}^2$) and multiplying the entire equation by $2\sigma^2$ to simplify the expression, resulting in $-(N-p) + \frac{y^T(I-H)y}{\sigma^2} = 0$.

3.1 Missing Data

The rule, not the exception,

1. leads to unbalanced data
2. leads to less precision in model estimation
3. Important to address validity of analyses. This depends on assumptions about the missing data mechanism.

Let Y be the full response data, the set of responses planned to be observed by the study design.

e.g., TLC - BLLs to be obtained at weeks 0, 1, 4, 6

e.g., AIDS study plan to collect CD4 cell counts each week for 12 weeks. CD4 cell count is not meaningful for a dead person. may redefine the full response Y as CD4 cell counts each week and while alive.

Partition $Y = (Y^{\text{obs}}, Y^{\text{MIS}})$

Let $R_{ij} = I(Y_{ij} \text{ is observed})$

$R = \{R_{ij}\}$ is data

The full outcome data are (Y, R)

The full outcome model is

$$f(Y, R \mid X; \theta)$$

$X =$ covariate

$\theta =$ parameters

$$= f(Y^{\text{obs}}, Y^{\text{mis}}, R \mid X; \theta)$$

The full response model

$$f(Y \mid X, \theta) = \sum_{r \in R} f(y, r \mid x, \theta)$$

It is useful to factor the full outcome (Y, R) model as

$$\begin{aligned} f(Y, R \mid X, \theta) &= f(R \mid Y, X, \theta) \cdot f(Y \mid X, \theta) \\ &= f(Y \mid R, X, \theta) \cdot f(R \mid X, \theta) \end{aligned}$$

> [!TIP] This factorization is fundamental in missing data analysis. The first factorization, $f(R \mid Y, X) f(Y \mid X)$, represents the **selection model** approach (modeling the probability of being observed given the outcome). The second, $f(Y \mid R, X) f(R \mid X)$, represents the **pattern-mixture model** approach (modeling the outcome distribution conditional on the missingness pattern).

selection model factorization "pattern mixture model" factorization

$$f(R|Y, X, \theta) \quad \text{missing data mechanism}$$

Three types of missing data mechanism:

1. MCAR

$$f(R|Y^{\text{obs}}, Y^{\text{MIS}}, X; \theta) = f(R|X; \theta)$$

The prob. that a planned response is missing depends on neither Y^{obs} NOR Y^{MIS}.

> [!TIP] **MCAR** stands for "Missing Completely At Random." When data are MCAR, the missingness mechanism is entirely independent of both observed and unobserved data, as shown by the fact that the probability distribution of the response indicator R depends only on the covariates X and parameters θ.

We plan to measure test scores on students for each of 2 years $\Rightarrow Y_1, Y_2$. In year 1, we get Y_1 on 1000 students.

In year 2, due to budget constraints, we randomly select 800 of the 1000 students from year 1 to get Y_2.

Even if our selection of the 800 students for the year 2 test score is related to a covariate, e.g. sex we take 2/3 girls, 1/3 boys to obtain Y_2.

3.2 2) MAR

$$f(R|Y^{obs}, Y^{mis}, X; \theta) = f(R|Y^{obs}, X; \theta)$$

Missingness depends on the observed response but not the missing response.

> [!TIP] This is the definition of **MAR (Missing at Random)**. It contrasts with: * **MCAR (Missing Completely at Random):** The probability of missingness is independent of all data. * **MNAR (Missing Not at Random):** The probability of missingness depends on the values of the missing data itself.

e.g. if students with lower values of Y_1 are less likely to take the second test, then whether Y_2 is observed depends on the value of Y_1.

(3) **MNAR** missing not at random or "nonignorable" e.g. drop-out in a smoky cessation study Y_j = smoking status at occasion $j, j = 1, 2$ $R_2 = 0$ if Y_2 is not observed.

$$P(R_2 = 0 \mid Y_1, Y_2 = 0)$$

$$\ominus P(R_2 = 0 \mid Y_1, Y_2 = 1)$$

> [!TIP] **Missing At Random (MAR)** is defined by the condition $P(R_2 \mid Y_1, Y_2) = P(R_2 \mid Y_1)$, meaning the missingness depends only on the observed data. If the probability of missingness depends on the unobserved value Y_2 (i.e., the two expressions above are unequal), the data is **Missing Not At Random (MNAR)**.

$\rightarrow$ if equality holds $\implies$ MAR if equality does not hold $\implies$ MNAR

Recall our outcomes are (Y, R).

$$\begin{aligned} f(Y, R \mid X; \theta) &= f(R \mid Y, X; \theta) f(Y \mid X; \theta) \\ &= f(R \mid Y^{\text{obs}}, Y^{\text{mis}}, X; \theta) f(Y^{\text{obs}}, Y^{\text{mis}} \mid X; \theta) \end{aligned}$$

MCAR:

$$f(R \mid Y^{\text{obs}}, Y^{\text{mis}}, X; \theta) = f(R \mid X; \theta)$$

$$f(Y, R \mid X; \theta) = f(R \mid X; \theta) f(Y^{\text{obs}}, Y^{\text{mis}} \mid X; \theta)$$

THE likelihood is

$$\begin{aligned} L(\theta) &= \int_{Y^{\text{mis}}} f(R \mid X; \theta) f(Y^{\text{obs}}, Y^{\text{mis}} \mid X; \theta) \, dy^{\text{mis}} \\ &= f(R \mid X; \theta) f(Y^{\text{obs}} \mid X; \theta) \end{aligned}$$

Suppose $\theta = (\theta_1, \theta_2)$ so that

> [!TIP] This parameter separation ($\theta = (\theta_1, \theta_2)$) is key to likelihood-based inference. When the parameters governing the missingness mechanism (θ_2) are distinct from those governing the data distribution (θ_1), the likelihood factorizes, simplifying the estimation process.

$$L(\theta_1, \theta_2) = f(R \mid X; \theta_2) f(Y^{\text{obs}} \mid X; \theta_1)$$

3.3 MAR

$$f(R \mid y^{\text{obs}}, y^{\text{mis}}, X; \theta) = f(R \mid y^{\text{obs}}, X; \theta)$$

> [!TIP] **Missing At Random (MAR)** is a condition where the probability of missingness depends only on the observed data (y^{obs}) and covariates (X), meaning the missingness can be ignored in likelihood-based inference if the parameters θ are distinct.

3.4 MNAR

> [!TIP] **Missing Not At Random (MNAR)** occurs when the probability of missingness depends on the missing values themselves (y^{mis}). Unlike MAR, the missingness mechanism is non-ignorable, and the distribution of the missing data cannot be recovered from the observed data without making explicit assumptions about the missingness mechanism.

Q: Value of ME model over fixed effects?

$$Y_i = X_i\beta + Z_i b_i + \epsilon_i$$

$n_i \times 1 \qquad n_i \times p \quad p \times 1 \qquad n_i \times q \quad q \times 1 \qquad n_i \times 1$

The term $Z_i b_i + \epsilon_i$ (circled in red) is labeled e_i.

Independ

- $b_i \sim N(0, G)$
- $\epsilon_i \sim N(0, R_i)$

Often $R_i = \sigma^2 I_{n_i}$

For subj. i:

$$\begin{aligned} E(Y_i \mid b_i) &= X_i\beta + Z_i b_i \\ E(Y_i) &= X_i\beta \\ \text{Var}(Y_i \mid b_i) &= R_i \end{aligned}$$

$$\Sigma_i = \text{Var}(Y_i) = Z_i G Z_i^T + R_i$$

$q \times q$ (annotated under $Z_i G Z_i^T$)

3.5 Two-Stage formulation

3.5.1 Stage I:

$$Y_i = Z_i\beta_i + \varepsilon_i$$

Typically Z_i contains a col. of 1s, so β_i contains an intercept, so other vars in Z_i must vary within subj.

3.5.2 Stage II:

$$\beta_i = A_i\beta + b_i$$

here A_i is a design matrix β are constants (fixed effects) b_i are random $b_i \sim N(0, G)$

$$\begin{aligned} E(\beta_i) &= A_i\beta \\ Var(\beta_i) &= G \end{aligned}$$

Substituting the Stage II model into the Stage I model:

$$\begin{aligned} Y_i &= Z_i(A_i\beta + b_i) + \varepsilon_i \\ &= Z_iA_i\beta + Z_ib_i + \varepsilon_i \\ &= X_i\beta + Z_ib_i + \varepsilon_i \end{aligned}$$

> [!TIP] The final equation, $Y_i = X_i\beta + Z_ib_i + \varepsilon_i$, is the standard form of a **Linear Mixed-Effects Model**. The writer implicitly defines the fixed-effects design matrix X_i as Z_iA_i, simplifying the expression by combining the design matrices.

3.6 Random Intercept

$$\begin{aligned} Y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_i + \epsilon_{ij} \\ &= (\beta_0 + b_i) + \beta_1 t_{ij} + \epsilon_{ij} \end{aligned}$$

$$b_i \sim N(0, \sigma_b^2) \quad R_i = \sigma^2 I$$

3.7 Random Intercept & Slope

$$\begin{aligned} Y_{ij} &= \beta_0 + \beta_1 t_{ij} + b_{i1} + b_{i2}t_{ij} + \epsilon_{ij} \\ &= (\beta_0 + b_{i1}) + (\beta_1 + b_{i2})t_{ij} + \epsilon_{ij} \end{aligned}$$

$$\begin{pmatrix} b_{i1} \\ b_{i2} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, G\right) \quad G = \begin{pmatrix} \sigma_{b1}^2 & \sigma_{b12} \\ \sigma_{b12} & \sigma_{b2}^2 \end{pmatrix}$$

$$\begin{aligned} &H_0 : \text{Rand. Int.} \\ &H_1 : \text{Rand Int and Slope} \end{aligned}$$

> [!TIP] These hypotheses (H_0 and H_1) typically correspond to a likelihood ratio test used for model selection. The null hypothesis (H_0) represents the simpler model with only a random intercept, while the alternative (H_1) represents the more complex model that includes both a random intercept and a random slope.

3.8 Prediction of b_i

$$\hat{b}_i = E(b_i|Y_i; \hat{\beta}, \hat{G}, \hat{R}_i)$$

$$f(b_i|Y_i) = \frac{f(Y_i|b_i)f(b_i)}{\int f(Y_i|b_i)f(b_i)\,db_i}$$

(posterior for b_i)

is mult. normal

$E(b_i|Y_i)$ **is the BLUP**

$$= \int b_i f(b_i|Y_i)\,db_i$$

$$= GZ_i^T\Sigma_i^{-1}(Y_i - X_i\beta)$$

$$\Sigma_i = Z_iGZ_i^T + R_i$$

empirical BLUP:

$$\hat{b}_i = \hat{G}Z_i^T\hat{\Sigma}_i^{-1}(Y_i - X_i\hat{\beta})$$

The predicted response profile for subj i

$$\begin{aligned}\hat{Y}_i &= X_i\hat{\beta} + Z_i\hat{b}_i \\ &= (\hat{R}_i\hat{\Sigma}_i^{-1})X_i\hat{\beta} + (I - \hat{R}_i\hat{\Sigma}_i^{-1})Y_i\end{aligned}$$

3.9 Homework: Exercise 8.1

here $R_i = \sigma^2 I$

Examine the estimated correlation of Y_i for the 2 models (random int, random int and slope)

4 Introduction

This is a book created from markdown and executable code.

See @knuth84 for additional discussion of literate programming.

```
1 + 1
```

```
[1] 2
```

5 Summary

In summary, this book has no content whatsoever.

```
1 + 1
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[1] 2
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References